COMPUTER NETWORKS (DU #2304CS301)

COMPLETE 1-HOUR EXAM REVISION STUDY GUIDE — UNITS 1 TO 5

EXAM PREPARATION GUIDE — PREPARED FOR TANISH UCHHADIYA
Syllabus Scope: 100% complete coverage of Units 1, 2, 3, 4, and 5 from all course lectures and slides.
Key Contents: Definitions, classifications, topologies, step-by-step solved delay & subnetting numericals, comparison matrices, layer-by-layer OSI/TCP-IP functions, device breakdowns, IPv4/IPv6 headers, DNS hierarchy, email flow, web hosting, practice question bank, and a 1-page quick cheat sheet.

Unit	Unit Topic & Focus Area	Core Syllabus Concepts Covered
Unit 1	Introduction to Computer Networks	Network Characteristics, Advantages/Disadvantages, Applications, LAN/MAN/WAN, Architectures (P2P vs Client-Server), Topologies (Bus, Star, Ring, Mesh, Tree, Hybrid), Delay/Loss/Throughput Formulas & Solved Numericals
Unit 2	Transmission Media & Wireless PAN	Guided Media (UTP/STP Categories, Coaxial Baseband/Broadband, Fiber Optic TIR & Layers), Unguided Media (Ground/Sky/LOS Propagation, Radio, Microwave, Infrared), Bluetooth Piconet vs Scatternet
Unit 3	Reference Models & Switching	Simplex/Half-Duplex/Full-Duplex Modes, Circuit Switching (3 Phases) vs Packet Switching (Store-Forward), Connection-Oriented vs Connectionless, 7-Layer OSI Model, 4/5-Layer TCP/IP Model & Protocol Suite
Unit 4	Networking Devices & Servers	NIC (MAC Address), Repeater, Hub (Passive/Active/Intelligent), Modem (DSL/ADSL), Switch (L2/L3, MAC Table), Bridge, Router (Routing Tables), Gateway, Access Point, Dedicated/Shared & Functional Servers
Unit 5	Protocols, IP Addressing & Web	Protocols & Packets, IPv4 Header (20-60B), IPv6 Header (40B), Classful Addressing (A-E), Subnetting & CIDR Calculation Methods & Exercises, WWW, Web Hosting Steps, HTTP (Persistent/Non-Persistent), Email (SMTP/POP3/IMAP), FTP, DNS, Applications
Section 6	Final Revision Sheet & Exam Prep	1-Page Formula & Conversion Cheat Sheet, Master Comparison Tables, 25 High-Yield Practice Questions (10 Descriptive, 5 Numericals, 10 MCQs) with Full Answer Key

UNIT 1: INTRODUCTION TO COMPUTER NETWORKS

Foundational Concepts, Characteristics, Classifications, Architectures, Topologies & Network Delay Calculations

1.1 What is a Computer Network?

Definition: A computer network is an interconnected collection of autonomous computing devices (computers, servers, smartphones, IoT nodes, printers) linked by physical wiring or wireless communication channels to exchange data, share hardware/software resources, and communicate under standardized protocols.

Real-World Network Analogies:

- **Road Transportation Network:** Vehicles carrying cargo travel over interconnected roads and switches, analogous to data packets traversing network links and routers.
- **Electrical Power Grid:** Central power generation stations distribute energy across transmission lines to residential and commercial nodes, akin to servers serving client requests.
- **Telephone System:** Connects subscribers globally using dedicated switching centers to facilitate two-way real-time communication.
- **Social Media & Human Networks:** Interlinked individuals sharing knowledge, messages, and digital resources.

1.2 Core Characteristics of Computer Networks

- **Resource Sharing:** Peripheral devices (printers, scanners), storage volumes, and application software are accessed simultaneously by multiple network users.
- **High Reliability & Redundancy:** Critical files and databases are replicated across alternate servers; if one node fails, duplicate copies remain available.

• **Cost Efficiency (Saving Money):** Centralized resource sharing reduces hardware capital expenditure. A single high-capacity server serves multiple client workstations.

• **Powerful Communication Medium:** Facilitates instant global messaging, real-time video conferencing, and multi-user collaborative document editing.

• **Scalability:** Network capacity and processing power can be expanded seamlessly by integrating additional switches, routers, or server nodes.

• **Flexibility of Equipment Location:** Devices can be moved within LAN boundaries without reconfiguring the underlying network architecture.

• **Improved System Performance:** Distributed computing allows heavy computational workloads to be partitioned across multiple nodes.

1.3 Advantages and Disadvantages of Computer Networks

Advantages: Streamlined communication, cost savings, centralized storage and management, remote data access, and shared internet connectivity.

Disadvantages: Vulnerability to cyber security breaches, rapid propagation of viruses/malware, server dependencies creating single points of failure, network congestion under peak loads, and administrative maintenance overhead.

1.4 Applications of Computer Networks

• **Business & Finance:** Corporate email, ERP systems, teleconferencing, and remote directory access.

• **Banking & Commerce:** Automated Teller Machines (ATMs), online banking portals, digital payment gateways, and e-commerce.

• **Healthcare:** Telemedicine, digital patient records (EHR), and remote medical consultation.

• **Education & Media:** Online e-learning platforms, virtual classrooms, and streaming video.

• **IoT & Automation:** Smart homes, industrial sensor telemetry, and automated manufacturing pipelines.

1.5 Network Classifications by Scale (LAN, MAN, WAN)

Comparison Parameter	Local Area Network (LAN)	Metropolitan Area Network (MAN)	Wide Area Network (WAN)
Geographical Scope	Small localized area (room, office, building, campus)	Metropolitan region or entire city	Country, continent, or global scale
Approximate Range	1 km to 10 km	10 km to 100 km	Beyond 100 km (Global)
Ownership Model	Private (Single organization/owner)	Private or Public utility	Private or Public (ISPs/Telcos)
Data Transmission Speed	High Speed (100 Mbps to 10 Gbps+)	Moderate Speed (10 Mbps to 1 Gbps)	Lower Speed (kbps to Mbps)
Propagation Delay	Very Short (Microseconds)	Moderate (Milliseconds)	Long Propagation Delay
Network Equipment	NICs, Switches, Hubs, Ethernet Cables	Modems, Fiber Backbones, Routers	Routers, Satellite Transponders, Microwaves

1.6 The Internet & Communication Protocols

The **Internet** is a global, public network of interconnected autonomous networks communicating via standardized protocols. A **Protocol** is a formal set of rules and conventions governing data exchange between networked devices.

Key Protocol Elements:

1. *Syntax:* Data format, field structures, and bit-level layout of messages.
2. *Semantics:* Meaning assigned to specific bit patterns and control fields.
3. *Timing:* Speed matching, sequencing, and session synchronization.

1.7 Computer Network Architectures

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• **Peer-to-Peer (P2P) Architecture:** Every node (peer) possesses equal capabilities and acts simultaneously as a client and a server. There is no dedicated central controller. Best suited for small networks (≤10 workstations).

- *Advantages:* Inexpensive (no server hardware required), simple setup, node failure does not collapse the network.
- *Disadvantages:* No centralized backup, weak decentralized security, performance slows dramatically under heavy traffic.

• **Client-Server Architecture:** Centralized model comprising a high-performance central host (Server) managing files, security, and directory databases, and distributed client workstations that request server tasks.

- *Advantages:* Centralized data backup, robust centralized security administration, high performance and scalability.
- *Disadvantages:* High initial setup cost, requires a specialized Network Operating System (NOS), server is a single point of failure.

1.8 Computer Network Topologies

Topology defines the physical or logical layout of nodes and communication links:

• **Bus Topology:** All nodes connect to a single central cable known as the *backbone cable*. Data transmitted by a node propagates in both directions; the destination node accepts the frame based on address matching. *Pros:* Minimal cabling, inexpensive. *Cons:* Backbone break downs entire network, signal interference/collisions, difficult fault isolation.

• **Ring Topology:** Every node connects to exactly two adjacent nodes, forming a closed circular loop. Data circulates in one direction via a *token passing* mechanism; repeaters regenerate signals. *Pros:* Equal network access, no packet collisions. *Cons:* Single node/cable failure disrupts entire ring, adding nodes introduces communication delay.

• **Star Topology:** Every node connects directly to a central multiport controller (Hub or Switch) via dedicated point-to-point links. All traffic passes through the central controller. *Pros:* Easy installation and troubleshooting, cable failure affects only the connected node. *Cons:* Central hub failure downs whole network, higher cable consumption than Bus.

• **Mesh Topology:** Every device is connected to every other device via dedicated point-to-point links. For a network with N nodes, the total number of physical duplex links required is: **L = N(N - 1) / 2**. Each node requires (N - 1) input/output ports. (e.g., 5 nodes = 5(4)/2 = **10 links**). *Pros:* Maximum fault tolerance, highly reliable, zero traffic congestion, dedicated security. *Cons:* Very expensive, complex wiring and installation.

• **Tree Topology:** Hierarchical arrangement where Star-configured sub-networks connect to a central backbone bus cable (parent-child hierarchy). Widely used in WANs and corporate campus backbones. Supports broadband long-distance transmission.

• **Hybrid Topology:** Integrates two or more distinct topologies (e.g., Star-Bus or Star-Ring) to combine their operational strengths.

Topology Name	Physical Arrangement	Dedicated Link Formula	Installation & Cost	Troubleshooting Ease
Bus	Single linear backbone cable	L = 1 (Shared cable)	Easy / Inexpensive	Difficult (Hard to find break)
Ring	Unidirectional closed loop	L = N (Point-to-point)	Easy / Inexpensive	Difficult (Breaks whole loop)
Star	Central Hub or Switch	L = N (Point-to-central)	Easy / Moderate Cost	Very Easy (Isolated ports)
Mesh	Full Point-to-Point connections	L = N(N - 1) / 2	Complex / Very Expensive	Easy (Dedicated links)
Tree	Hierarchical Parent-Child Tree	L = N - 1	Complex / High Cost	Easy (Segmented)
Hybrid	Combination of 2+ topologies	Varies by structure	Complex / High Cost	Difficult

1.9 Network Performance Metrics & Solved Numericals

When a packet travels from source to destination, it incurs four distinct nodal delay components:

Total Nodal Delay Formula:
d_nodal = d_proc + d_queue + d_tran + d_prop

1. *Processing Delay (d_proc):* Time host/router spends examining packet headers, processing bit errors, and determining output interface.
2. *Queuing Delay (d_queue):* Time packet waits in router buffer queues prior to transmission. Depends on traffic intensity.
3. *Transmission Delay (d_tran):* Time required to push all packet bits onto the link.
   Formula: d_tran = L / R (where L = Packet length in bits, R = Transmission rate/Bandwidth in bps).

4. *Propagation Delay (d_prop):* Time required for a bit to physically travel over link distance at wave speed.
Formula: d_prop = d / s (where d = Distance in meters, s = Propagation speed in m/s, typically ~2x10^8 m/s).

SOLVED EXERCISE 1 — TRANSMISSION & PROPAGATION DELAY CALCULATION
Problem: A 2,000-byte packet is sent over a 1,500 km link with a transmission rate of 100 Mbps. Propagation speed s = 2.5x10^8 m/s. Calculate d_tran and d_prop.
Solution:
• Packet Size L = 2,000 Bytes x 8 bits/Byte = **16,000 bits**.
• Transmission Rate R = 100 Mbps = 100 x 10^6 bps = 10^8 bps.
• **d_tran = L / R** = 16,000 / 10^8 = 0.00016 seconds = **0.16 ms**.
• Distance d = 1,500 km = 1,500,000 meters = 1.5 x 10^6 m.
• **d_prop = d / s** = (1.5 x 10^6) / (2.5 x 10^8) = 15 / 2,500 = 0.006 seconds = **6.0 ms**.

SOLVED EXERCISE 2 — TOTAL NODAL DELAY COMPREHENSIVE PROBLEM
Problem: A packet of 1,200 bytes is sent across a 1,500 km link at 500 Mbps. Speed s = 2.5x10^8 m/s. Average queuing delay d_queue = 4 ms, processing delay d_proc = 1.5 ms. Find total nodal delay.
Step 1 (d_tran): L = 1,200 x 8 = 9,600 bits. R = 500 x 10^6 bps. d_tran = 9,600 / 5x10^8 = 0.0000192 s = **0.0192 ms**.
Step 2 (d_prop): d = 1,500,000 m. d_prop = 1,500,000 / 2.5x10^8 = 0.006 s = **6.0 ms**.
Step 3 (d_nodal): d_nodal = 1.5 ms (proc) + 4.0 ms (queue) + 0.0192 ms (tran) + 6.0 ms (prop) = **11.5192 ms**.

Packet Loss & Throughput Calculations:

• **Packet Loss:** Occurs when router buffers overflow during heavy traffic. Formula: Packets Lost = Total Transmitted x Loss Rate.
Example: 1,500 packets sent with 5% loss rate -> Lost = 1,500 x 0.05 = **75 packets lost**.

• **Throughput:** Actual rate of successful data delivery over a channel.
Formula: Throughput = Data Successfully Received / Time Taken.
Example: A 2 GB file transfer receives 1.8 GB successfully in 45 seconds.
Data in MB = 1.8 x 1024 MB = 1,843.2 MB. Throughput = 1,843.2 / 45 = **40.96 MB/s**.

UNIT 2: TRANSMISSION MEDIA & WIRELESS NETWORKING

Guided (Wired) Pathways, Unguided (Wireless) Media, Wave Propagation Modes & Bluetooth WPAN

2.1 Introduction & Design Factors

Transmission Media represents the physical or wireless channel carrying signals from Source to Destination at Layer 1 (Physical Layer).

Key Design Factors & Impairments:

1. *Bandwidth:* Maximum data carrying capacity of a channel in bits per second (bps, Kbps, Mbps, Gbps).
2. *Transmission Impairments:* Signal degradation during propagation due to:
- **Attenuation:** Loss of signal energy/amplitude over distance (requires amplifiers/repeaters).
- **Distortion:** Shape alteration of composite signals due to varying component frequencies.
- **Noise / Crosstalk / EMI:** Unwanted external electrical signals combining with the original signal.

2.2 Guided Transmission Media (Wired)

Guided media provides physical conductors along which signals travel:

• **Twisted-Pair Cable:** Consists of pairs of insulated copper wires twisted in a continuous helix to cancel out Electromagnetic Interference (EMI) and crosstalk.

- *Unshielded Twisted Pair (UTP):* Lacks metallic shielding. Lightweight, flexible, low cost. Standard Ethernet categories: Cat 1 (Voice), Cat 2 (4 Mbps), Cat 3 (10 Mbps), Cat 4 (16 Mbps), Cat 5 (100 Mbps), Cat 5e (1 Gbps), Cat 6 (10 Gbps), Cat 7 (10 Gbps+).

- *Shielded Twisted Pair (STP):* Wrapped in conductive metal foil/braid shield. Provides superior noise immunity for outdoor/industrial environments but requires proper electrical grounding.

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• **Coaxial Cable:** Constructed with 4 concentric layers: Inner Copper Conductor, Dielectric Insulator, Outer Braided Copper Shield, Protective Outer Jacket.

- *Transmission Modes:* Baseband (single digital channel over full bandwidth, e.g., LANs, CCTV) vs Broadband (multiple analog channels using FDM, e.g., Cable TV).

- *Connectors:* BNC (Bayonet connector), BNC-T (T-connector for node tap), BNC Barrel (cable coupler), 50-Ohm BNC Terminator (prevents wave reflection in Bus networks).

• **Fiber Optic Cable:** Transmits high-speed light pulses using **Total Internal Reflection** inside glass/plastic threads. 5 Layers: Core (high refractive index n1), Cladding (lower index n2), Buffer Coating, Kevlar Strength Member, Outer Jacket.

- *Types:* Single-Mode Fiber (narrow core, laser source, ultra long-distance WANs) vs Multi-Mode Fiber (wider core, LED source, short-distance LANs). Connectors: SC, ST, MT-RJ.

Media Type	Internal Construction	Bandwidth Capacity	EMI / Noise Immunity	Cost & Max Distance
UTP Cable	4 twisted copper pairs (unshielded)	Up to 10 Gbps (Cat 6/7)	Low (Susceptible to EMI)	Lowest Cost / 100 meters
STP Cable	Twisted copper with foil shield	Up to 10 Gbps+	High (Foil noise barrier)	Moderate Cost / 100 meters
Coaxial Cable	Solid copper + Dielectric + Braid	High (Baseband & Broadband)	High (Outer braid shield)	Moderate Cost / ~500 meters
Fiber Optic	Glass Core (n1) + Cladding (n2)	Ultra High (Gbps to Tbps)	Total (Immune to EMI/Noise)	Highest Cost / 10 to 100+ km

2.3 Unguided Transmission Media (Wireless)

Unguided media transmits electromagnetic signals through air or free space without physical conductors.

• Atmospheric Wave Propagation Modes:

1. *Ground Wave Propagation (<2 MHz):* Low-frequency signals hug the Earth's curvature. Used for AM radio broadcasting and maritime navigation.

2. *Sky Wave Propagation (2 to 30 MHz):* Signals bounce off the ionized layer of the atmosphere (ionosphere). Used for long-distance shortwave international radio.

3. *Line-of-Sight (LOS) Propagation (>30 MHz):* High-frequency signals travel in straight direct lines between aligned antennas. Used for FM radio, TV, satellite.

• Wireless Spectrum Categories:

- **Radio Waves (3 kHz to 1 GHz):** Omnidirectional, penetrate walls, no antenna alignment needed. Applications: AM/FM radio, TV, cellular networks.

- **Microwaves (1 GHz to 300 GHz):** Unidirectional focused beams, strict LOS required, blocked by obstacles.
* *Terrestrial Microwave:* Ground antennas (4 to 23 GHz) spaced 30-50 km apart.
* *Satellite Microwave:* Geostationary transponder relay (1 to 40 GHz). Uses **Uplink** (earth-to-satellite) and **Downlink** (satellite-to-earth) frequencies.

- **Infrared Waves (300 GHz to 400 THz):** Short-range LOS, blocked by solid walls (confined to room). Applications: TV remotes, wireless mice, IR security sensors.

Property	Radio Waves	Microwaves	Infrared Waves
Frequency Range	3 kHz to 1 GHz	1 GHz to 300 GHz	300 GHz to 400 THz
Propagation Direction	Omnidirectional (spreads everywhere)	Unidirectional (focused line beam)	Line-of-Sight (short range)
Wall Penetration	Yes (penetrates walls easily)	No (reflected/blocked by walls)	No (strictly confined to room)
Primary Applications	AM/FM Radio, TV, Cordless Phones	Satellite TV, Wi-Fi, Cellular, Radar	TV Remotes, Wireless Mouse/Keyboard

2.4 Bluetooth Technology & Wireless PAN

Bluetooth is a short-range wireless Personal Area Network (WPAN) operating in the **2.4 GHz ISM band** (2.400 - 2.4835 GHz). Uses **Frequency-Hopping Spread Spectrum (FHSS)** across 79 channels at 1,600 hops/sec to prevent interference.

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• **Piconet:** Master-slave architecture containing 1 Master node and up to **7 Active Slave nodes** (max 8 active nodes total). Up to 255 parked nodes can remain synchronized. Direct slave-to-slave communication is prohibited.

• **Scatternet:** Formed by interconnecting multiple Piconets via **Bridge Nodes** (a node acting as Master in one Piconet and Slave in another).

Comparison Aspect	Piconet Architecture	Scatternet Architecture
Topology	Single master controlling up to 7 active slaves	Multiple overlapping interconnected piconets
Bridge Element	Not applicable	Bridge Node (connects separate piconets)
Active Device Capacity	Maximum 8 active devices (1 Master + 7 Slaves)	Scalable across multiple piconet clusters

UNIT 3: REFERENCE MODELS & NETWORK SWITCHING

Communication Modes, Switching Techniques, Connection Approaches, 7-Layer OSI & 4/5-Layer TCP/IP Models

3.1 Mode of Communication

Feature	Simplex Mode	Half-Duplex Mode	Full-Duplex Mode
Directionality	Unidirectional (1-way only)	Bidirectional (2-way, 1 at a time)	Bidirectional (Simultaneous 2-way)
Channel Utilization	Full bandwidth used for 1 direction	Full bandwidth alternates direction	Bandwidth split across 2 channels
Device Roles	Fixed Sender and Receiver	Both transmit, but alternately	Both transmit & receive simultaneously
Real-World Example	Keyboard to PC, TV Broadcast	Walkie-Talkie	Telephone Call, Video Conference

3.2 Network Switching Techniques

• **Circuit Switching:** Establishes a dedicated physical circuit between source and destination before transmission.
- *3 Operational Phases:* Connection Establishment -> Data Transfer -> Circuit Disconnect.
- *Characteristics:* Guaranteed bandwidth, zero queuing delay during transfer, no store-and-forward. Used in voice telephony.

• **Packet Switching:** Messages are divided into discrete packets carrying source/destination addresses. Packets travel independently over shared links using **Store-and-Forward** at routers. Packets may arrive out-of-order and are reassembled at destination. Used in Internet data networks.

Comparison Parameter	Circuit Switching	Packet Switching
Connection Path	Dedicated physical path established upfront	Dynamic independent routing per packet
Operational Phases	3 Phases (Setup, Transfer, Release)	No prior setup phase required
Store-and-Forward	No store-and-forward (continuous stream)	Uses Store-and-Forward at every router
Bandwidth Efficiency	Reserved bandwidth (wasted if idle)	Dynamic bandwidth sharing (highly efficient)
Primary Application	Voice Telephone Networks	Internet Data Communication

3.3 Connection-Oriented vs Connectionless Approach

• **Connection-Oriented:** Requires connection establishment phase. Guarantees in-order packet delivery, error checking, flow control, and acknowledgments. Example: **TCP**.

• **Connectionless:** Sends datagrams directly without connection setup. Fast and lightweight, but offers no delivery guarantees or packet order tracking. Example: **UDP**.

3.4 OSI Reference Model (ISO 1984 — 7 Layers)

Mnemonic to remember layers (Bottom-to-Top): **P**lease **D**o **N**ot **T**hrow **S**am's **P**izza **A**way.

Layer & Name	PDU Unit	Core Functions & Responsibilities	Associated Hardware & Protocols
L7 Application	Data	Provides network services directly to end-user application software.	Web Browsers / HTTP, HTTPS, FTP, SMTP, DNS

Layer & Name	PDU Unit	Core Functions & Responsibilities	Associated Hardware & Protocols
L6 Presentation	Data	Data translation (ASCII/EBCDIC), Encryption/Decryption, and Compression.	SSL/TLS, Compression Engines
L5 Session	Data	Establishes, manages, and terminates sessions; controls dialog (half/full duplex); adds sync checkpoints.	NetBIOS, Session Managers
L4 Transport	Segment / Datagram	End-to-end message delivery, Port addressing, Segmentation/Reassembly, Flow & Error Control.	Heart of OSI / TCP (Ports 80, 443, 21), UDP
L3 Network	Packet	Logical addressing (Source/Dest IP), Routing best path selection, Packetizing.	Routers, L3 Switches / IPv4, IPv6, ICMP, ARP
L2 Data Link	Frame	Hop-to-hop frame delivery, MAC addressing, LLC & MAC sublayers, Error detection (CRC).	Switches, Bridges, NIC Drivers / Ethernet, Wi-Fi
L1 Physical	Bit	Transmits raw bit stream over physical medium; signaling, bit synchronization, line configuration.	Hubs, Repeaters, Cables, Modems, NIC Hardware

3.5 TCP/IP Reference Model & Protocol Suite

Developed by US Department of Defense (DoD). Original 4 layers: Application, Transport, Internet, Link. Updated 5-layer model splits Link into Data Link and Physical layers.

OSI Model Layers (7)	TCP/IP Updated Model (5)	Core Protocols in TCP/IP Suite
Application, Presentation, Session	Application Layer	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, SNMP, Telnet
Transport Layer	Transport Layer	TCP (Reliable, Connection-oriented), UDP (Fast, Connectionless)
Network Layer	Internet / Network Layer	IP (IPv4/IPv6), ICMP (Control messages), IGMP, ARP
Data Link Layer	Data Link Layer	Ethernet (802.3), Wi-Fi (802.11), MAC addressing
Physical Layer	Physical Layer	Copper Cables, Fiber Optics, Radio Waves

UNIT 4: NETWORKING CONNECTING DEVICES & SERVERS

Layer 1 to Layer 7 Connecting Hardware, MAC vs IP Forwarding, Dedicated & Shared Server Types

4.1 Network Connecting Devices

• **Network Interface Card (NIC / LAN Card):** Hardware adapter providing network connectivity. Contains a unique burned-in **48-bit (6-byte) MAC Address** in hexadecimal (e.g., 06:02:03:04:4C:2B). Operates at Layer 1 & Layer 2.

• **Repeater:** Operates at Layer 1. Receives degraded signals over long distances, regenerates/amplifies them, and retransmits across 2 ports.

• **Hub:** Multiport repeater at Layer 1. Connects devices in a Star topology. **Broadcasts** all incoming signals to all ports (non-intelligent). Shares a single collision domain, operates in half-duplex. Types: Passive, Active (amplifies), Intelligent (monitored).

• **Modem (Modulator-Demodulator):** Converts digital computer data into analog signals for telephone/cable lines, and vice versa. Types: Cable, DSL, ADSL (Asymmetric DSL: higher download speed than upload speed).

• **Switch:** Multiport bridge operating at Layer 2 (Data Link Layer). Builds a **MAC Address Table** by learning source MACs. Forwards frames directly to destination port (unicast). Floods unknown destination MACs. Each port forms a separate collision domain.

• **Bridge:** Layer 2 device connecting LAN segments. Inspects MAC addresses to filter or forward frames, reducing traffic collision domains.

• **Router:** Layer 3 (Network Layer) internetworking device. Reads IP addresses, maintains **Routing Tables**, and forwards packets across different networks using optimal path selection (RIP, OSPF). Connects LANs to WANs.

• **Gateway:** Connects two completely dissimilar networks using different protocols. Acts as a **Protocol Converter** at Layer 7 (Application Layer).

• **Wireless Access Point (WAP):** Bridges wired Ethernet LAN and wireless Wi-Fi clients, creating a Wireless LAN (WLAN).

Connecting Device	OSI Layer	Addressing Used	Primary Function & Operational Logic
Repeater	Layer 1 (Physical)	None	Regenerates and amplifies signals across 2 network segments.
Hub	Layer 1 (Physical)	None	Non-intelligent; broadcasts incoming data frames to ALL connected ports.
Modem	Layer 1 (Physical)	None	Modulates digital-to-analog and demodulates analog-to-digital signals.
Bridge	Layer 2 (Data Link)	MAC Address	Filters and forwards frames between 2 LAN segments based on MAC table.
Switch	Layer 2 (Data Link)	MAC Address	Intelligent multiport bridge; forwards frames to target port using MAC Table.
Router	Layer 3 (Network)	IP Address	Connects different networks; routes IP packets using optimal routing tables.
Gateway	Layer 7 (Application)	Protocol Translation	Protocol converter enabling communication between dissimilar networks.
Access Point (WAP)	Layer 1 & 2	MAC Address	Bridges wired Ethernet and wireless Wi-Fi clients (creates WLAN).

4.2 Types of Computer Servers

A **Server** provides central resources, data, or services to Client systems over a network.

• **By Resource Allocation:** Dedicated Server (100% resources allocated to 1 user/org) vs Non-Dedicated / Shared Server (resources shared among multiple users).

• By Server Function:

Server Type	Core Functionality & Operational Purpose	Key Protocols & Examples
File Server	Centralized file storage, file retrieval, and user access management.	FTP, SMB, NFS
Application Server	Hosts and executes business application logic between client and database.	Java App Servers, PHP, Python WSGI
Print Server	Manages network printer access, print queues, and spooling.	LPD, IPP, SMB
Mail Server	Sends, receives, stores, and routes electronic mail messages.	SMTP (sending), POP3 / IMAP (retrieving)
Proxy Server	Intermediary between client and web; handles request forwarding, caching, & anonymity.	HTTP, HTTPS, SOCKS
Web Server	Stores website files (HTML, CSS, Images) and serves them to browsers.	HTTP, HTTPS, Apache, Nginx

UNIT 5: NETWORK PROTOCOLS, IP ADDRESSING & WEB SERVICES

IPv4 & IPv6 Header Formats, Classful IP Addressing, CIDR Subnetting, Web Hosting, HTTP, Email & DNS

5.1 Protocols & Packet Structure

A **Protocol** defines communication rules (What, How, When). Categories: *Management* (SNMP, ICMP), *Communication* (TCP, UDP, IP, HTTP), *Security* (HTTPS, TLS, SSH). A **Packet** consists of a **Header** (control/address info) and **Payload** (actual user data).

5.2 Internet Protocol (IP) & Header Formats

• **IPv4 Header Format:** Minimum **20 bytes** (up to 60 bytes with Options). Key fields: *Version* (4 bits), *HLEN* (4 bits, header length in 4-byte words), *TOS* (8 bits, Quality of Service), *Total Length* (16 bits), *Identification / Flags (DF, MF) / Fragment Offset* (for packet fragmentation), *TTL (Time to Live)* (hop count limit), *Protocol* (8 bits, 6 = TCP, 17 = UDP), *Header Checksum* (16 bits error check), *Source IP* (32 bits), *Destination IP* (32 bits).

• **IPv6 Header Format:** Fixed **40 bytes** (320 bits) header size. Key fields: *Version* (4 bits), *Traffic Class* (8 bits), *Flow Label* (20 bits), *Payload Length* (16 bits), *Next Header* (8 bits), *Hop Limit* (8 bits = TTL), *Source Address* (128 bits), *Destination Address* (128 bits).

Feature	IPv4 (Internet Protocol v4)	IPv6 (Internet Protocol v6)
Address Length	32 bits (4 bytes)	128 bits (16 bytes)
Address Representation	Dotted Decimal (e.g., 192.168.1.10)	8 Hexadecimal groups (e.g., 2001:0db8::8a2e:0370:7334)
Header Size	Variable (20 to 60 bytes)	Fixed 40 bytes

Feature	IPv4 (Internet Protocol v4)	IPv6 (Internet Protocol v6)
Header Checksum	Present (recalculated at every router hop)	Removed (handled by Layer 2 and Transport Layer)
Total Address Space	2^32 (~4.3 Billion addresses)	2^128 (~3.4 x 10^38 addresses)

5.3 IP Addressing & Classful Classification

An IPv4 address is a 32-bit binary number represented in dotted-decimal format (4 octets). Divided into **Net ID** (identifies network) and **Host ID** (identifies specific host).

Class	First Byte Range	Prefix Bits	Net ID / Host ID Bits	Default Subnet Mask	Purpose / Capacity
Class A	0 to 127	0...	8 Bits Net ID / 24 Bits Host ID	255.0.0.0 (/8)	Huge networks (16.7M hosts)
Class B	128 to 191	10...	16 Bits Net ID / 16 Bits Host ID	255.255.0.0 (/16)	Medium networks (65,534 hosts)
Class C	192 to 223	110...	21 Bits Net ID / 8 Bits Host ID	255.255.255.0 (/24)	Small networks (254 hosts)
Class D	224 to 239	1110...	Multicast Address Group	None	Multicast Communication
Class E	240 to 255	1111...	Reserved / Experimental	None	Research & Future Use

• **Special Addresses:** *Network Address* (first address, host bits all 0s), *Broadcast Address* (last address, host bits all 1s). **Usable Hosts per Network = 2^H - 2**.

5.4 Subnetting & CIDR (Classless Inter-Domain Routing)

Subnetting splits a network into smaller subnets by borrowing host bits for Net ID.
• **Number of Subnets created:** 2^N (where N = borrowed host bits).
• **Total Addresses per Subnet:** 2^H (where H = remaining host bits).
• **Usable Hosts per Subnet:** 2^H - 2.
• **CIDR Notation:** IP_Address / Prefix_Length (e.g., 192.168.1.0/26 means 26 network bits, 6 host bits).

> **SOLVED SUBNETTING EXERCISE — CIDR 192.168.1.0 /26**
> **Given:** Class C IP 192.168.1.0/26. Default Class C mask = /24.
> **Step 1 (N Borrowed Bits):** N = 26 - 24 = **2 bits**. Number of subnets = 2^2 = **4 subnets**.
> **Step 2 (New Mask):** 24 ones + 2 ones = 11111111.11111111.11111111.11000000 = **255.255.255.192**.
> **Step 3 (Hosts per Subnet):** Remaining host bits H = 32 - 26 = 6 bits. Total = 2^6 = 64. Usable = 64 - 2 = **62 hosts**.
> **Subnet Ranges:** Subnet 0: 192.168.1.0 to .63 | Subnet 1: .64 to .127 | Subnet 2: .128 to .191 | Subnet 3: .192 to .255.

5.5 Web Technologies & Application Protocols

• **World Wide Web (WWW):** Global repository of web pages linked by hypertext, written in HTML, identified by URLs, and transferred via HTTP/HTTPS.

• **Web Hosting Workflow Steps:** *Domain Registration -> Choose Hosting Provider -> Server Configuration -> File Upload (FTP) -> Database Setup -> DNS Setup -> Live Site & Maintenance*. Types: Shared, VPS, Cloud, Dedicated, Reseller.

• **HTTP (HyperText Transfer Protocol):** Client-server protocol on Port 80 (HTTPS on 443). Stateless protocol. *Non-Persistent HTTP:* Closes TCP connection after each request/response. *Persistent HTTP:* Keeps TCP connection open for multiple requests.

• **Email System & Protocols:** Components: Mail User Agent (MUA), Message Transfer Agent (MTA), Message Queue, Mailbox.
- *SMTP (Port 25):* Pushes emails from client to server and server to server.
- *POP3 (Port 110):* Downloads emails from server to client device and deletes from server.
- *IMAP (Port 143):* Syncs emails across devices while retaining messages on server.

• **FTP (File Transfer Protocol):** Transfers files using 2 separate TCP connections: **Control Connection (Port 21)** for commands and **Data Connection (Port 20)** for file transfer.

• **DNS (Domain Name System):** Translates domain names (e.g., www.google.com) into IP addresses (e.g., 173.194.36.51). Hierarchy: **Root DNS Servers (13 global) -> TLD Servers (.com) -> Authoritative Servers -> Local Resolver Cache**.

SECTION 6: FINAL 1-HOUR REVISION SHEET & EXAM PREPARATION

Master Formula Cheat Sheet, Key Comparisons, 25 High-Yield Exam Questions & Complete Answer Key

6.1 Master Formula & Unit Conversion Cheat Sheet

Metric / Calculation	Formula / Derivation	Unit Conversions & Exam Tips
Transmission Delay	d_tran = L / R	L in bits (1 Byte = 8 bits); R in bps (1 Mbps = 10^6 bps).
Propagation Delay	d_prop = d / s	d in meters (1 km = 1,000 m); s in m/s (typically 2x10^8 or 2.5x10^8 m/s).
Total Nodal Delay	d_nodal = d_proc + d_queue + d_tran + d_prop	Sum all 4 sub-delays in identical time units (ms or s).
Mesh Topology Links	L = N(N - 1) / 2	N = Number of nodes. Every node has (N-1) dedicated duplex links.
Subnet Calculations	Subnets = 2^N | Usable Hosts = 2^H - 2	N = Borrowed host bits; H = Remaining host bits (32 - CIDR prefix).
Throughput	Throughput = Data Received / Time	Data in Bytes/bits; Time in seconds (1 GB = 1024 MB).

6.2 High-Yield Exam Question Bank (25 Questions)

Part A: Descriptive Model Questions

Q1: Define a Computer Network. Discuss its core characteristics and key sector applications. [Unit 1]

Q2: Compare LAN, MAN, and WAN across scale, ownership, delay, and equipment. [Unit 1]

Q3: Differentiate between Peer-to-Peer and Client-Server network architectures with diagrams. [Unit 1]

Q4: Compare Bus, Star, Ring, and Mesh topologies. Derive the Mesh link formula for N nodes. [Unit 1]

Q5: Differentiate UTP and STP cables. List Ethernet UTP categories from Cat 1 to Cat 7. [Unit 2]

Q6: Explain Fiber Optic cable layers and the principle of Total Internal Reflection. [Unit 2]

Q7: Compare Radio Waves, Microwaves, and Infrared Waves in unguided media. [Unit 2]

Q8: Describe Bluetooth Piconet and Scatternet architectures and active device limits. [Unit 2]

Q9: Explain the 7 layers of the OSI model with PDUs and core functions of each layer. [Unit 3]

Q10: Compare Hub, Switch, Router, Bridge, and Gateway with respect to OSI layers. [Unit 4]

Part B: Solved Exam Numericals

Q11: Calculate transmission delay for a 2,000-byte packet over a 100 Mbps link.
Solution: L = 2000 x 8 = 16,000 bits. R = 100x10^6 bps. d_tran = 16,000 / 10^8 = 0.00016 s = **0.16 ms**.

Q12: Calculate propagation delay between two nodes 1,500 km apart (s = 2.5x10^8 m/s).
Solution: d = 1,500,000 m. d_prop = 1,500,000 / (2.5x10^8) = 0.006 s = **6.0 ms**.

Q13: A network loses 2% of packets. If 1,000 packets are sent, how many are lost?
Solution: Lost Packets = 1,000 x 0.02 = **20 packets**.

Q14: Find the class, default mask, network address, and broadcast address for IP 185.28.17.9.
Solution: Class B (128-191). Default Mask: 255.255.0.0 (/16). Network: **185.28.0.0**. Broadcast: **185.28.255.255**.

Q15: Calculate the number of subnets and usable hosts per subnet for IP 139.10.10.1 /20.
Solution: Default Class B = /16. Borrowed N = 20 - 16 = 4 bits. Subnets = 2^4 = **16 subnets**. Host bits H = 32 - 20 = 12. Usable Hosts = 2^12 - 2 = **4,094 hosts**.

Part C: Practice Multiple Choice Questions (MCQs)

Q16: Which topology requires a dedicated point-to-point physical link between every pair of nodes?
(A) Star (B) Mesh (C) Bus (D) Ring

Tanish Uchhadiya

Q17: What is the maximum number of active slave devices in a Bluetooth Piconet?
(A) 255 (B) 8 (C) 7 (D) 15

Q18: At which OSI layer does a Layer 2 Switch primarily operate?
(A) Physical (B) Data Link (C) Network (D) Transport

Q19: Which protocol is used by email clients to SEND emails to a mail server?
(A) POP3 (B) IMAP (C) SMTP (D) FTP

Q20: What is the fixed header size of an IPv6 packet?
(A) 20 Bytes (B) 60 Bytes (C) 40 Bytes (D) 128 Bytes

Q21: Which connector is used at both ends of a main coaxial cable in Bus topology?
(A) SC Connector (B) RJ-45 (C) 50-Ohm BNC Terminator (D) MT-RJ

Q22: In the OSI model, which layer is known as the 'Heart of OSI'?
(A) Network (B) Transport (C) Data Link (D) Application

Q23: Which wave propagation mode reflects off the ionosphere for shortwave broadcasting?
(A) Ground Wave (B) Sky Wave (C) Line-of-Sight (D) Surface Wave

Q24: FTP uses which port for establishing its CONTROL connection?
(A) Port 20 (B) Port 21 (C) Port 25 (D) Port 80

Q25: How many bits are present in a standard hardware MAC address?
(A) 32 bits (B) 64 bits (C) 48 bits (D) 128 bits

ANSWER KEY FOR PRACTICE MCQS (Q16 TO Q25)
Q16: B (Mesh) — Mesh uses dedicated point-to-point links between all nodes.
Q17: C (7) — Piconet supports 1 Master + 7 Active Slaves (8 total active devices).
Q18: B (Data Link Layer) — Switches operate at Layer 2 using MAC addresses.
Q19: C (SMTP) — Simple Mail Transfer Protocol is used for pushing/sending emails.
Q20: C (40 Bytes) — IPv6 basic header has a fixed size of 40 bytes (320 bits).
Q21: C (50-Ohm BNC Terminator) — Absorbs reflected signals at cable ends.
Q22: B (Transport Layer) — Transport layer bridges upper software and lower hardware layers.
Q23: B (Sky Wave) — Sky wave reflects off the ionosphere in the 2-30 MHz range.
Q24: B (Port 21) — FTP uses Port 21 for Control and Port 20 for Data connection.
Q25: C (48 bits / 6 Bytes) — Standard MAC address is 48 bits long represented in hexadecimal.